

ADITYA TIWARI

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Profile Summary

Results-oriented **ML & Platform Engineer** with nearly 4 years of experience delivering highly reliable AI solutions and overseeing robust **infrastructure operations**. I specialize in bridging advanced **predictive modeling** with rigorous **platform management**, ensuring machine learning systems are seamlessly deployed, scaled, and continuously maintained. My expertise spans the complete **AI lifecycle**, from driving **zero-downtime platform migrations** and ensuring cluster reliability, to architecting next-generation **Generative AI** solutions, including sophisticated **Retrieval-Augmented Generation (RAG)** systems. I am dedicated to championing **operational stability** and **production excellence** for complex, data-driven applications.

Experience

Data Scientist

Jio Platforms Limited

Mumbai, India

July 2022 – Present

- **End-to-End Platform Ownership:** Manage the internal JDSP Platform for the organization, taking full responsibility for development, cluster management, and maintaining a high-availability environment 99.9% for 500+ users.
- **Infrastructure Migration & Stabilization:** Successfully executed a zero-downtime migration of legacy Airflow and JDSP workloads, stabilizing core infrastructure and leading complex user workspace transitions across environments (e.g., HDP to CDP).
- **Containerization & Orchestration:** Orchestrate containerized services using Kubernetes and Docker, managing liveness probes and troubleshooting complex pod lifecycle issues (Exit Code 137) to ensure system resilience.
- **Systems & OS Management:** Handle comprehensive on-premise server activities, including OS-level maintenance, cluster upgrades, scaling operations, and infrastructure optimization to support data-intensive workloads.
- **Automation & CI/CD:** Streamline operational efficiency by automating manual processes and maintaining robust deployment pipelines through Git/GitHub, ensuring reliable version control and platform growth.
- **Expert Technical Support:** Provide dedicated L1/L2 end-to-end support, troubleshooting platform-specific bottlenecks and delivering fast resolution for data engineering and data science teams.
- **Advanced Workflow Integration:** Lead feature enhancements to streamline workflows, integrating modern tools for data orchestration with Apache Airflow and preparing the platform for scalable AI and RAG-based application deployments.

Projects

Stateful AI Travel Agent (LangGraph, Gemini, Streamlit)

[Project Report](#) 

- Architected a stateful AI agent using **LangGraph** and **LangChain**, powered by the **Gemini** LLM, to autonomously reason and generate dynamic, personalized travel itineraries.
- Integrated the **Tavily** Search API as a custom tool within the agentic workflow, enabling real-time data retrieval and reducing AI hallucinations for location-specific queries.
- Developed an interactive front-end using **Streamlit** to facilitate seamless human-AI interaction, structuring the backend for clean, scalable local deployment.

Technical Skills

- **Programming / Scripting:** Python, SQL, Bash Scripting
- **ML Frameworks:** PyTorch, TensorFlow
- **Generative AI & LLMs:** LangChain, Hugging Face, RAG, Gemini API, LangGraph
- **MLOps & Infrastructure:** Apache Airflow, Kubernetes (K8s), Docker, GitLab, Prometheus, Grafana
- **Data Analysis & Backend:** Pandas, NumPy, FastAPI, Streamlit

Education

Global Nature Care Sangathan's Group Of Institution, Jabalpur

BTech in Electronics And Telecommunication

July 2018 – July 2022

- CGPA: 8.35/10.0

Certifications

- **Machine Learning Specialization** – Stanford Online / DeepLearning.AI (2025)
- **Generative AI with Large Language Models Specialization** – DeepLearning.AI (2026)
- **Python for Everybody Specialization** – University of Michigan (2024)